

THEORY AND METHODS

PERTURBATION OF THE NORMAL MODEL
IN LINEAR REGRESSION

D.J. de Waal, Department of Mathematical Statistics, University of
P.C.N. Groenewald the Orange Free State, P.O. Box 339, Bloemfontein,
and C.J. Kemp: 9300 Republic of South Africa.

Key words: Generalized extreme value distribution; linear model; nonnormal errors; perturbation; reference priors.

Summary: The linear model is one of the most commonly used inferential tools in statistics where it is assumed that the errors are distributed $N(0, \sigma^2)$. The assumption of log-normal distributed errors are often made so that the standard application of normal theory can be made after transformation. The log-normal distribution is, however, a special case of a class of generalized Gamma distributions of which the Weibull distribution is also a member. After logarithmic transformation follows the class of generalized extreme value distributions with a parameter k that reduces to a type I extreme value distribution (corresponding to the Weibull distribution) when $k = 1$ and to a normal distribution when $k \rightarrow \infty$. In this paper perturbation of the normal model in linear regression to a class of generalized extreme value (GEV) distributions is considered. A reference prior under the GEV model is used and the posterior distributions of the parameters of the linear model are derived for two sets of data by using the Gibbs sampler. Perturbation diagnostics were used on the predictive densities of some parameters.

1. INTRODUCTION

The linear model $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \epsilon$ is one of the most commonly used inferential tools in statistics where it is assumed that the x_i 's are fixed and ϵ is distributed $N(0, \sigma^2)$. It follows that Y conditional on $\underline{x}' = (1 \ x_1 \dots x_p)$

is distributed $N(\underline{x}'\underline{\beta}, \sigma^2)$, where $\underline{\beta} = (\beta_0 \beta_1 \dots \beta_p)'$.

The assumption that ε , the error, is normally distributed can, however, be questioned in many applications, especially in lifetime data. In such cases the assumption of log-normally distributed errors is often made so that the standard normal theory can be applied after transformation. The log-normal distribution is, however, a special case of a class of generalized Gamma distributions of which the Weibull distribution is also a member. Thus, after logarithmic transformation, the variable may be modelled by a class of generalized extreme value (GEV) distributions with a parameter k that reduces to a type I extreme value (EV) distribution when $k = 1$ (corresponding to the Weibull distribution) and to a normal distribution when $k \rightarrow \infty$.

Grummer-Strawn (1993) analyzed breastfeeding data by assuming that the time of breastfeeding, $Z|X = x$, follows a Weibull distribution with the location parameter of the form $\exp(-\underline{x}'\underline{\beta})$. To be more specific, for $k = 1$, $Z \sim \text{Weibull}(\exp(-\underline{x}'\underline{\beta} + \gamma\tau), 1/\tau)$, where γ is the Euler constant and τ is a precision parameter, and $Y = -\ln Z \sim \text{EV}(\underline{x}'\underline{\beta} - \gamma\tau, \tau)$ with $E[Y|X = x] = \underline{x}'\underline{\beta}$ and variance $\text{Var}[Y|X = x] = \frac{\pi^2}{6\tau^2}$.

Another application of the linear model where Y conditional on covariates X is not normally distributed, is in estimation of process variation (Grego, 1993) where $Y = \ln S^2$ is expressed in terms of a linear function of factor level settings. (S^2 denotes cell variances.) This problem is also addressed by Atkinson, 1982 and McCullagh and Nelder, 1989.

Using the framework of perturbation analysis (Geisser, 1993), one can view the model as a class of generalized extreme value distributions with k as perturbation parameter and the normal distribution as standard, i.e. when $k \rightarrow \infty$.

2. THE GENERALIZED EXTREME VALUE DISTRIBUTION

The generalized extreme value (GEV) distribution that we will consider is the following (Patil, Boswell and Ratnaparki, 1984):

The variable ε is distributed GEV $(\sqrt{k}\tau\psi(k), \tau, k)$ if its density function is given by

$$(2.1) \quad f(\varepsilon|k) = \frac{1}{\sqrt{k}\Gamma(k)\tau} \exp \left[-\frac{\sqrt{k}}{\tau} (\varepsilon - \sqrt{k}\tau\psi(k)) - \exp \left(-(\varepsilon - \sqrt{k}\tau\psi(k))/\sqrt{k}\tau \right) \right],$$

$$-\infty < \varepsilon < \infty$$

$$\tau > 0$$

$$k > 0,$$

$$\text{where } \psi(k) = \frac{d}{dk} \ln \Gamma(k)$$

$$= -\gamma - \frac{1}{k} + \lim_{n \rightarrow \infty} \sum_{j=1}^n \left[\frac{1}{j} - \frac{1}{k+j} \right],$$

and $\gamma = 0.5772157\dots$ is Euler's constant.

It can be shown that

$$E(\varepsilon) = 0 \text{ and } \text{Var}(\varepsilon) = k\tau^2\zeta(2, k),$$

where $\zeta(r, q) = \sum_{n=0}^{\infty} 1/(q+n)^r$ is Riemann's Zeta function (Gradshteyn and

Ryzhik, 1980). It also follows that $\zeta(2, k) = \psi'(k)$, the trigamma function.

If $k = 1$ in (2.1), the distribution of ε becomes the extreme value (EV) distribution with density function

$$(2.2) \quad f(\varepsilon|k=1) = \frac{1}{\tau} \exp \left[-\frac{1}{\tau}(\varepsilon + \gamma\tau) - \exp(-(\varepsilon + \gamma\tau)/\tau) \right],$$

$$-\infty < \varepsilon < \infty,$$

mean $E(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \frac{\pi^2\tau^2}{6}$. It will be denoted by $\varepsilon \sim EV(-\gamma\tau, \tau)$.

The EV distribution function is given by

$$(2.3) \quad F(\varepsilon) = \exp \left[-\exp(-(\varepsilon + \gamma\tau)/\tau) \right],$$

which is useful for simulation purposes, since an observation can be generated by

$$\varepsilon = -\tau \ln(-\ln g) - \gamma\tau,$$

where g is a random $(0, 1)$ number.

To simulate a GEV-value, use the result (Patil, Boswell, Ratnaparki, 1984, p. 61) that if $W \sim \text{Gamma}(k, 1)$, then $\varepsilon = -\sqrt{k}\tau (\ln W + \psi(k)) \sim \text{GEV}(\sqrt{k}\tau \psi(k), \tau, k)$.

$$(2.4) \quad \text{Hence } \varepsilon = \sqrt{k}\tau \psi(k) - \sqrt{k}\tau \ln \left[\sum_{j=1}^k (-\ln g_j) \right]$$

where $g_j, j = 1, \dots, k$, is a set of random $(0, 1)$ numbers. Note that (2.4) only applies for k an integer. The distribution function can also be written as

$$(2.5) \quad F(\varepsilon) = 1 - \frac{1}{\Gamma(k)} \int_0^{\exp(\psi(k) - \varepsilon/\sqrt{k}\tau)} w^{k-1} e^{-w} dw.$$

Figure 2.1 presents the EV distribution ($k = 1$), the GEV distributions for $k = 0.5$ and 2 , given $\tau = 0.2$, and the normal $N(0, \tau^2 = 0.2^2)$ distribution ($k \rightarrow \infty$).

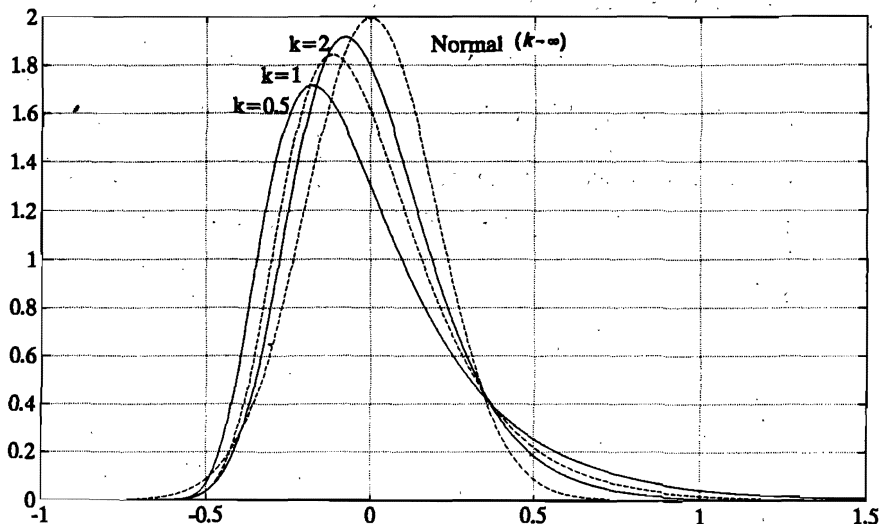


Figure 2.1. GEV distribution for $k = 0.5; 1; 2; \infty$ and $\tau = 0.2$.

Remarks:

- (i) The generalized extreme value distribution is also a generalized Gompertz distribution (see Patil, Boswell and Ratnaparki, 1984).
- (ii) For ε distributed as in (2.1), $Z = \exp(-\varepsilon)$ is distributed as a generalized Gamma distribution with density function given by

$$(2.6) \quad f(z|k) = \frac{k^{k-\frac{1}{2}}}{\Gamma(k)\tau} \mu^{-\sqrt{k}/\tau} z^{\sqrt{k}/\tau-1} e^{-(kz/\mu)^{1/\sqrt{k}\tau}}, \quad z > 0$$

where $\mu = e^{-\sqrt{k}\tau \psi(k)}$

The r -th moment of Z about the origin is

$$E(Z^r) = \mu^r k^{r/\sqrt{k}\tau} \Gamma(k + r\tau/\sqrt{k})/\Gamma(k).$$

3. BAYESIAN ESTIMATION AND PREDICTION

For

$$Y = \underline{x}' \underline{\beta} + \varepsilon$$

and ε distributed as in (2.1), Y has the GEV distribution with

$$E[Y | \underline{\beta}, \tau, k, \underline{x}] = \underline{x}' \underline{\beta} \quad \text{and variance}$$

$$\text{Var}[Y | \underline{\beta}, \tau, k, \underline{x}] = k \tau^2 \zeta(2, k).$$

Here the posterior of $\underline{\beta}$ and τ will be derived for fixed k , as well as the predictive density of a future observation.

- (i) $k \rightarrow \infty$: Normal case.

When $k \rightarrow \infty$, the distribution of ε approaches a $N(0, \tau^2)$ distribution, and $Y | \underline{\beta}, \tau, \underline{x} \sim N(\underline{x}' \underline{\beta}, \tau^2)$.

$$\left[\lim_{k \rightarrow \infty} k \zeta(2, k) = 1, \text{ see Gradshteyn and Ryzhik (1980)} \right].$$

This model leads to the standard normal analysis, and with the noninformative prior

$$(3.1) \quad p(\underline{\beta}, \tau) \propto \frac{1}{\tau}$$

on the parameters, it follows (de Groot, 1970) that the marginal posterior distributions of $\underline{\beta}$ and τ are given by

$$(3.2) \quad \underline{\beta} | \underline{y}, X \sim t_p(n-p, \hat{\underline{\beta}}, (n-p)s^2(X'X)^{-1}),$$

a multivariate t distribution with $n-p$ degrees of freedom, location parameter $\hat{\underline{\beta}} = (X'X)^{-1}X'\underline{y}$ where X is $n \times p$, and precision matrix $(s^2)^{-1}X'X$, with

$$s^2 = \frac{1}{n-p} (\underline{y} - X\hat{\underline{\beta}})'(\underline{y} - X\hat{\underline{\beta}}),$$

and

$$(3.3) \quad \tau^{-2} | \underline{y}, X \sim \text{Gamma}\left(\frac{1}{2}(n-p+3), \frac{1}{2}(n-p)s^2\right).$$

The prior given in (3.1) is the reference prior for this problem (Bernardo, 1979, Berger and Bernardo, 1992). The invariant Jeffreys prior (Jeffreys, 1961) is

$$(3.4) \quad p(\underline{\beta}, \tau) \propto \frac{1}{\tau^{p+1}},$$

but it is not generally felt (even by Jeffreys) to be the most suitable for a location-scale problem (Berger, 1985), and (3.1) was in standard use even before the advent of the reference prior.

The Bayes estimates of $\underline{\beta}$ and τ under squared error loss are the expectations

$$E[\underline{\beta} | \underline{y}, X] = \hat{\underline{\beta}} \quad \text{and} \quad E[\tau | \underline{y}, X] = \frac{\Gamma[\frac{1}{2}(n-p+2)]}{\Gamma[\frac{1}{2}(n-p+3)]} \left(\frac{1}{2}(n-p)s^2\right)^{\frac{1}{2}}.$$

The predictive density of a future observation Y_{n+1} given $\underline{x}_0$ is

$$(3.5) \quad Y_{n+1} | (\underline{y}, X, \underline{x}_0) \sim t_1(n-p, \underline{x}_0' \hat{\underline{\beta}}, (n-p)s^2(1 + \underline{x}_0'(X'X)^{-1}\underline{x}_0)).$$

(ii) k finite: General extreme value case.

Under the error distribution (2.1), it follows that

$$Y | \underline{\beta}, \tau, k, \underline{x} \sim \text{GEV}(\underline{x}'\underline{\beta} + \sqrt{k}\tau\psi(k), \tau, k).$$

Fisher's information matrix for fixed k is the $(p+1) \times (p+1)$ matrix

$$I = \begin{bmatrix} I(\underline{\beta}, \underline{\beta}') & I(\underline{\beta}, \tau) \\ I(\tau, \underline{\beta}') & I(\tau, \tau) \end{bmatrix}$$

where

$$(3.6) \quad \begin{aligned} I(\underline{\beta}, \underline{\beta}') &= \frac{1}{\tau^2} X' X, \\ I(\underline{\beta}, \tau) &= -\frac{n}{\sqrt{k} \tau^2} \bar{x}, \\ I(\tau, \tau) &= \frac{n}{\tau^2} [1 + k\psi'(k)] \end{aligned}$$

$$\text{and } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

It follows then directly that the Jeffreys prior and the reference prior for $\underline{\beta}$ and τ are again given by (3.4) and (3.1) respectively.

The joint posterior distribution of $\underline{\beta}$ and τ given y, X, k under a joint prior $p(\underline{\beta}, \tau) \propto \frac{1}{\tau}$ (reference prior) or $p(\underline{\beta}, \tau) \propto \frac{1}{\tau^{p+1}}$ (Jeffreys prior) is given by

$$(3.7) \quad \begin{aligned} p(\underline{\beta}, \tau | y, X, k) &\propto f(y | X, k, \underline{\beta}, \tau) p(\underline{\beta}, \tau) \\ &\propto \tau^{-1} \exp \left[-\frac{\sqrt{k}}{\tau} \sum_{i=1}^n (y_i - \underline{x}_i \underline{\beta} - \sqrt{k} \tau \psi(k)) \right. \\ &\quad \left. - \sum_{i=1}^n \exp \left(-(y_i - \underline{x}_i \underline{\beta} - \sqrt{k} \tau \psi(k)) / \sqrt{k} \tau \right) \right] \\ &\quad \times p(\underline{\beta}, \tau). \end{aligned}$$

An explicit form for this posterior distribution is not available, but by using the Gibbs sampler (see, for example, Gelfand and Smith, 1990, Casella and

George, 1992) for a specific data set, quite accurate marginal posteriors for the β 's and τ can be obtained. These posteriors are shown for a few cases considered in section 5. The predictive density of a future Y_{n+1} given x_0 is also not available in closed form, but from each set of simulated posterior (β, τ) 's as obtained through the Gibbs sampler the density $f(y_{n+1}|x_0, \beta, \tau)$ can be calculated. Repeating this procedure for all simulated sets of posterior (β, τ) 's and averaging the densities, we obtained a smoothed predictive density for Y_{n+1} as shown in section 5.

4. PERTURBATION OF THE NORMAL

It was mentioned earlier that if $k \rightarrow \infty$ the GEV model becomes the normal model. To consider the effect of perturbing the normal to the GEV model, we will look at the predictive density of Y_{n+1} for a given x_0 for various values of the perturbation parameter k . Let $f_0(y_{n+1}|x_0)$ denote the predictive density of Y_{n+1} under the normal model and $f_k(y_{n+1}|x_0)$ the predictive density under the GEV model.

Various measures to quantify the effect of the perturbation have been proposed in the literature (Geisser, 1992). One is the comparison of the average predictive loss. If $L(a, y_{n+1})$ is the loss incurred when action a is taken and a future realization y_{n+1} is obtained, then the average predictive loss is

$$\bar{L}_k(a) = \int L(a, y_{n+1}) f_k(y_{n+1}|x_0) dy_{n+1}$$

which is minimized at the optimal action a_k^* . If a_0^* is the optimal action under the standard model, the differential loss is

$$d(k) = \bar{L}_k(a_0^*) - \bar{L}_k(a_k^*) \geq 0,$$

and $\max_k d(k)$ measures the maximum possible effect of the perturbation.

Then

$$(4.1) \quad R(k) = \frac{d(k)}{\bar{L}_k(a_k^*)}$$

measures the relative increase in risk by using the standard.

For quadratic loss when predicting a future observation, $L(a, y_{n+1}) = (a - y_{n+1})^2$, it follows that

$$(4.2) \quad R(k) = (E_0[Y_{n+1} | x_0] - E_k[Y_{n+1} | x_0])^2 / \text{Var}_k[Y_{n+1} | x_0].$$

Another measure (Johnson and Geisser, 1982, 1983, Geisser, 1992), is the Kullback-Leibler divergence applied to the predictive densities,

$$(4.3) \quad K(k, \infty) = E[\ell_n f_k - \ell_n f_0],$$

where expectation is over the perturbed distribution. This measure is useful in indicating the effect of the perturbation on the predictive density, but there may be some difficulty in the exact interpretation (Geisser, 1992) of $\max_k K(k, \infty)$.

Other measures of the distance between densities, such as the L^1 norm (Devroye, 1987), or the Pitman (1979) norm are special cases of the Hellinger distance raised to the m -th power,

$$(4.4) \quad H^m = \int |f_0^{1/m} - f_k^{1/m}|^m dy.$$

Another readily interpretable measure of the effect of perturbation is also proposed by Geisser (1992). This measure assesses the robustness of a $(1 - \alpha)$ highest posterior density (HPD) region based on the standard relative to a perturbed distribution. It can be done either by comparing the volumes of the respective HPD regions, or by examining the probability $1 - \alpha_k$ according to the perturbed distribution, for the $(1 - \alpha)$ HPD region of the standard.

The above measures can of course also be applied to any of the marginal posterior densities of the parameters instead of to the predictive densities.

Remark: If the original observations $z_1, \dots, z_n$ are from a Weibull distribution or from any of the generalized Gamma distributions, then we may want to predict Z_{n+1} instead of Y_{n+1} . The predictive density of Z_{n+1} follows from the transformation $Z_{n+1} = \exp(-Y_{n+1})$.

5. ILLUSTRATIONS

The above perturbations will be illustrated on two sets of data. The first is on the Gesell data (Mickey, Dunn and Clark, 1967) where Z is the age at first word and X is Gesell's adaptive score for a child. Let $Y = -\ln Z$, then the linear model $Y = \beta_0 + \beta_1 X + \varepsilon$ is the assumed model. The second set of data is the time Z a Farr sailboat (11,2 metres in length) takes to finish a course of length X_2 (in sea miles) under certain averaged wind conditions X_1 (in knots). Again we let $Y = -\ln Z$ and fit the model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$.

Example 1

Table 5.1: Gesell data: z_i is age in months at first word and x_i is Gesell's adaptive score of i -th child.

i	z_i	x_i	i	z_i	x_i	i	z_i	x_i
1	15	95	8	11	100	15	11	102
2	26	71	9	8	104	16	10	100
3	10	83	10	20	94	17	12	105
4	9	91	11	7	113	18	42	57
5	15	102	12	9	96	19	17	121
6	20	87	13	10	83	20	11	86
7	18	93	14	11	84	21	10	100

Let $y = -\ln z$ and fit the linear model $y = \beta_0 + \beta_1 x$. Least squares estimates of β_0 , β_1 and τ are given by $\hat{\beta}_0 = -4.1912$, $\hat{\beta}_1 = 0.0174$, $\hat{\tau} = 0.3661$.

Figure 5.1 shows a plot of y_i against x_i .

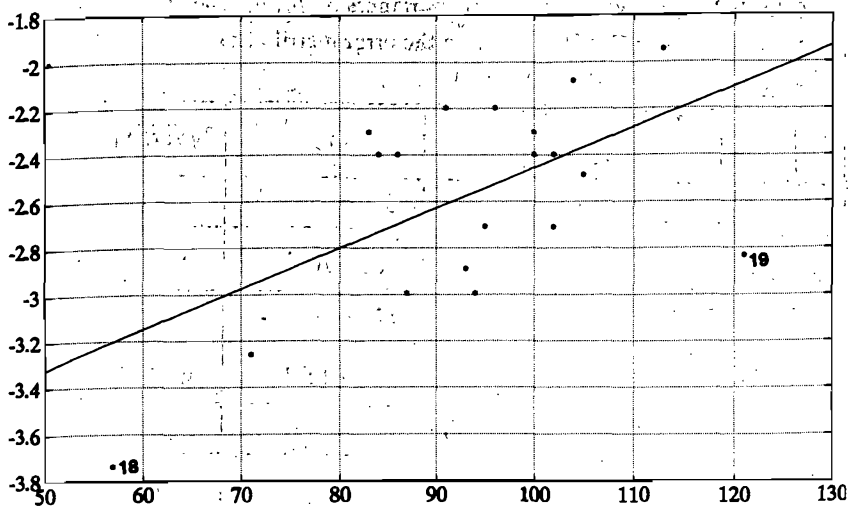


Figure 5.1. Plot of Gesell data and least squares fit

This data set is well-known and has been used by a number of authors to illustrate outlier detection approaches. (Chalenor and Brant, 1988, Pettit, 1992, Geisser, 1992, Van der Merwe and Botha, 1993). The general consensus is that observation 19 is an outlier while 18 is an influential observation. In all the above-mentioned analyses a normal model was assumed.

Performing the Bayes analysis described in section 3 on this data set for various values of k and with the reference prior (3.1), the marginal posteriors of the parameters were obtained using the Gibbs sampler on (3.7). Using an initial burn-in period of 50 simulations, samples of 2000 from the marginal posteriors were generated. In table 5.2 the posterior means and variances of the parameters are given for a few values of k . The last column gives the Bayes estimate of the standard deviation of the observations, $\tau\sqrt{k\zeta(2, k)}$.

Table 5.2. Posterior means and variances of β_0 , β_1 and τ for Gesell data under the error distribution $\text{GEV}(\sqrt{k}\tau \psi(k), \tau, k)$

k		β_0	β_1	τ	$\sqrt{\text{Var}(Y)}$
1	E	-4.085	0.0164	0.371	
	Var	0.0798	0.85×10^{-5}	3.33×10^{-3}	0.4793
2	E	-4.065	0.0161	0.380	
	Var	0.1002	1.09×10^{-5}	3.61×10^{-3}	0.4320
3	E	-4.128	0.0167	0.382	
	Var	0.1084	1.19×10^{-5}	3.67×10^{-3}	0.4154
4	E	-4.115	0.0166	0.382	
	Var	0.1228	1.35×10^{-5}	3.72×10^{-3}	0.4066
5	E	-4.235	0.0178	0.382	
	Var	0.1335	1.48×10^{-5}	3.75×10^{-3}	0.4010
∞	E	-4.191	0.0174	0.352	
	Var	0.3431	3.83×10^{-5}	3.14×10^{-3}	0.3524

From the table we notice that the perturbation from the normal does not affect the Bayes estimates of the regression parameters β_0 and β_1 very much. There is, however, a substantial reduction in the variances of the marginal posterior distributions, which seem to be increasing functions of k , while the standard deviation of the errors is a decreasing function of k .

As the main interest in this data set lies in determining if point 19 is an outlier, let us look at the predictive density for point 19, that is, for $\mathbf{x}_0 = (1 \ 121)$.

Figure 5.2 shows the predictive density of Y_{n+1} for $k=1$, i.e. the extreme value model, and for the normal model according to (3.5). The 95% prediction intervals are also indicated on the graph. In both cases the observed value of $Y, y = -2.833$, is included in the interval.

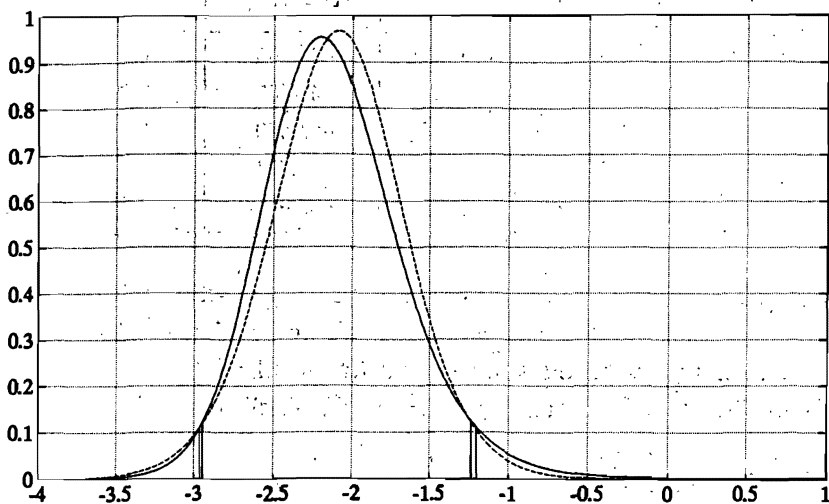


Figure 5.2. Predictive density of Y given $x_0 = (1 \ 121)$ for extreme value model (—) and the normal model (---)

The respective means, variances and prediction intervals are $E[Y | x_0] = -2.0885$, $\text{Var}[Y | x_0] = 0.1855$, $C = (-2.947; -1.237)$ for the normal model, and $E[Y | x_0] = -2.0685$, $\text{Var}[Y | x_0] = 0.2735$, $C = (-2.983; -0.967)$ for the extreme value model.

The 95% prediction interval translates in terms of the original measurements to (3.44; 19.05) for the normal model and (2.63; 19.74) for the extreme value model.

In table 5.3 the measures defined in (4.2)–(4.4) are given for a few

values of k together with the mean and variance of the corresponding predictive density.

Table 5.4. Perturbation measures and moments of predictive density

k	$R(k)$	$K(k, \infty)$	H^2	$E[Y_{n+1} x_0]$	$\text{Var}[Y_{n+1} x_0]$
0.5	0.0023	0.1715	0.0601	-2.0604	0.3393
1	0.0014	0.0791	0.0299	-2.0685	0.2735
5	0.00017	0.0127	0.0058	-2.0909	0.2037
8	4.4×10^{-5}	0.0067	0.0033	-2.0913	0.1923
14	7×10^{-6}	0.0036	0.0018	-2.0895	0.1911

As expected, all the measures are decreasing functions of k and maximum perturbation occurs when $k \rightarrow 0$. From the values of $R(k)$ it can be seen that the relative loss would be quite low when predicting a future observation using the standard instead of a GEV distribution with $k > 0.5$.

Example 2 Farr data

The data in the appendix consists of the time Z it took a Farr sailboat to finish a race of length X_2 (measured in sea miles) in an averaged wind speed during the race of X_1 (measured in knots). Here $n = 66$ finish-times have been recorded for three yachts. Not all three yachts sailed every race.

To be able to predict the time it will take a yacht to finish a course of a certain distance under a given wind strength, the model $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ will be used, where $Y = -\ln Z$.

A least squares fit gives estimates $\hat{\beta}_0 = -0.8825$, $\hat{\beta}_1 = 0.0428$, $\hat{\beta}_2 = -0.0527$ with $R^2_{Y,Y} = 0.8455^2 = 71\%$. The maximum likelihood estimate of τ under the assumption that $\varepsilon \sim N(0, \tau^2)$ is $\hat{\tau} = 0.2303$. A histogram (figure 5.3) of $\hat{\varepsilon} = y - \hat{y}$, where $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$, indicates, however, that ε may not be normally distributed.

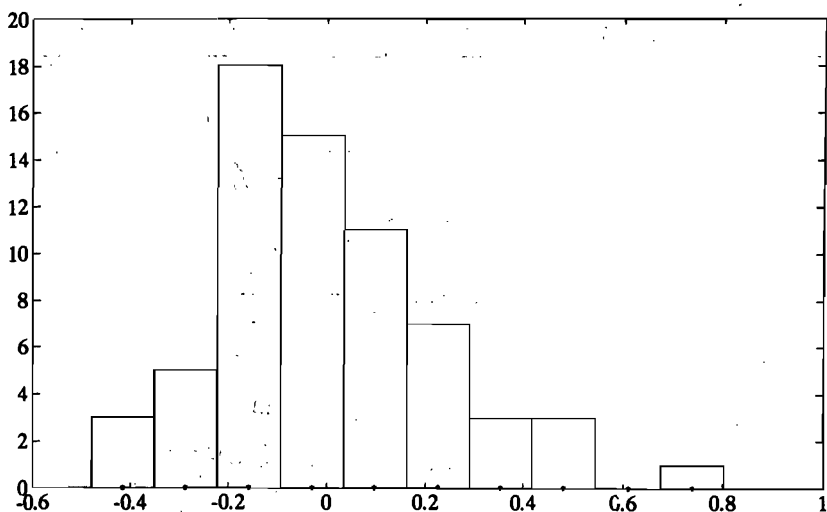


Figure 5.3. Histogram of estimated errors for the Farr data using least squares fit

If we consider the class of generalized extreme value distributions as perturbations from the normal, it seems that the case $k = 2$ may be a good candidate. A comparison of the empirical cumulative distribution function of errors e_i , $i = 1, \dots, 66$ with the cdf of a $\text{GEV}(\sqrt{k}\tau\psi(k), \tau, k)$ distribution for various integer values of k yielded a minimum sum of absolute deviations, $\sum |F - \hat{F}|$, when $k = 2$ (see figure 5.4). The parameter τ is taken as the Bayes estimate in table 5.4.

If we assume that $Y|\underline{\beta}, \tau, k, \underline{x} \sim \text{GEV}(\underline{x}'\underline{\beta} + \sqrt{k}\tau\psi(k), \tau, k)$, and the reference prior (3.1), the following Bayes estimates of $\underline{\beta}$ and τ were obtained after applying the same Gibbs sampling procedure as in example 1 to obtain the marginal posterior distributions.

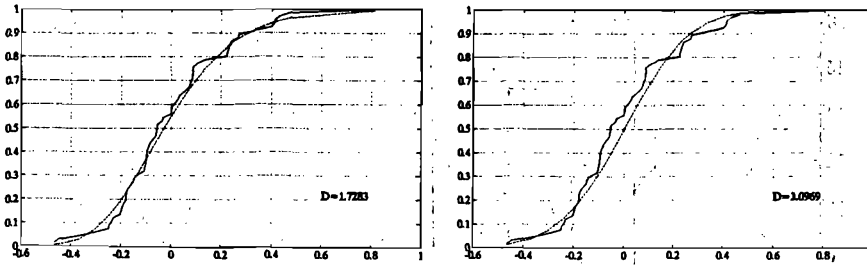


Figure 5.4. Empirical CDF of the errors (—) with
 (a) the CDF of a $GEV(\sqrt{k} \tau \psi(k), \tau, k)$ distribution for $k = 2$,
 (b) the CDF of a $N(0, \tau^2)$ distribution, where $D = \sum_{i=1}^{66} |F - \hat{F}|$

Table 5.4. Bayes estimates of $\beta_0, \beta_1, \beta_2, \tau$ and $\sqrt{\text{Var}(Y)}$ for different values of k

k	β_0	β_1	β_2	τ	$\sqrt{\text{Var}(Y)}$
1	-0.9473	0.0439	-0.0481	0.1993	0.2556
2	-0.9313	0.0439	-0.0498	0.2080	0.2362
3	-0.9174	0.0437	-0.0508	0.2125	0.2312
4	-0.9194	0.0436	-0.0507	0.2152	0.2293
∞	-0.8825	0.0428	-0.0527	0.2276	0.2276

Figure 5.5 shows the posterior distributions of $\beta_0, \beta_1, \beta_2$ and τ for $k=2$ under the GEV model compared with the posteriors under the normal model.

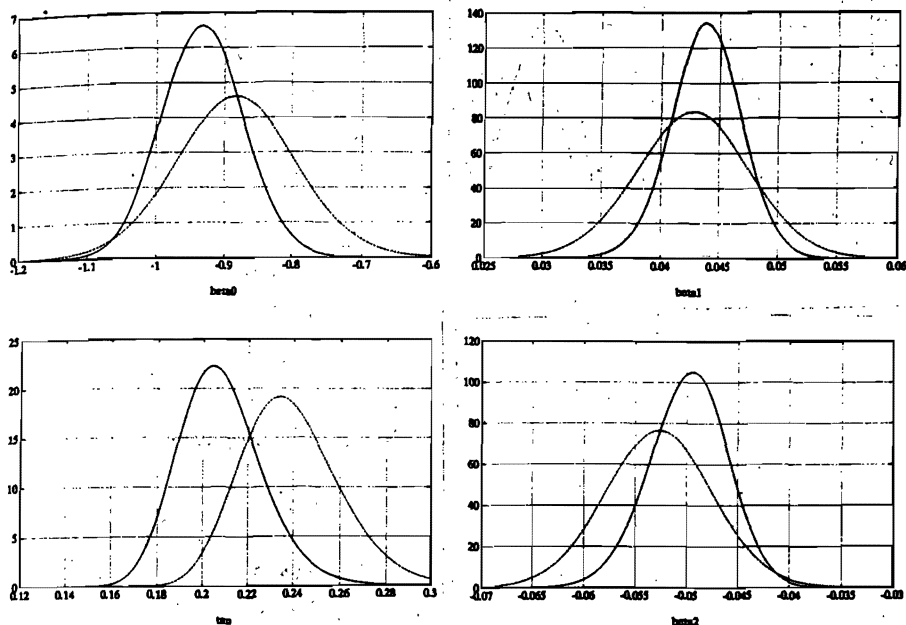


Figure 5.5. Posterior distribution of the parameters under the GEV model with $k=2$ (—) and under the normal model (---)

Observation 29 in the appendix looks like a potential outlier. The predictive density of Y_{n+1} given $\mathbf{z}_0 = (1 \ 15 \ 8)$ is shown in figure 5.6 for the normal model and for the GEV model with $k=2$. The observed value is $y = -0.1409$.

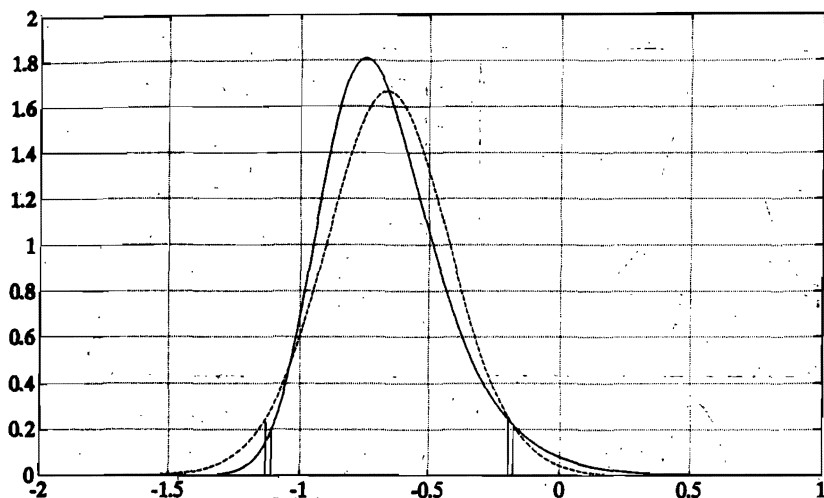


Figure 5.6. Predictive density of Y_{n+1} given $x_0 = (1 \ 15 \ 8)$ under GEV model with $k = 2$ (—) and under the normal model (---)

The respective means and variances are

$$E[Y_{n+1} | x_0, k = 2] = -0.6676, \text{ Var}[Y_{n+1} | x_0, k = 2] = 0.0592$$

$$E[Y_{n+1} | x_0, k = \infty] = -0.6617, \text{ Var}[Y_{n+1} | x_0, k = \infty] = 0.0587.$$

The 95% HPD intervals are

$$C = (-1.106; -0.176) \text{ for } k = 2, \text{ and} \\ C = (-1.130; -0.197) \text{ for } k = \infty.$$

In terms of the original observations, it follows after transformation that

$$E[Z_{n+1} | x_0, k = 2] = 2.0045, \text{ Var}[Z_{n+1} | x_0, k = 2] = 0.2056$$

$$E[Z_{n+1} | x_0, k = \infty] = 1.9950, \text{ Var}[Z_{n+1} | x_0, k = \infty] = 0.2206$$

with 95% prediction intervals

$$C = \begin{cases} (1.192 ; 3.022) & \text{for } k = 2, \text{ and} \\ (1.218 ; 3.096) & \text{for } k = \infty. \end{cases}$$

The Kullback-Leibler measure of divergence between the two densities is $K(2, \infty) = 0.042$ and the measure of relative quadratic loss, $R(2) = 0.00057$.

6. INFERENCES ABOUT k

In the previous sections k was considered a perturbation parameter and all inferences about the other parameters were made conditional on k . The amount of perturbation from the standard is no indication of the feasibility of an alternative model. So when the measures of divergence are unbounded it is necessary to examine these measures over a subset of the most likely values of the perturbation parameter.

In this section the reference prior distribution for all the parameters is derived where k is the parameter of interest. Fisher's information matrix is the $(p+2) \times (p+2)$ matrix

$$(6.1) \quad I(k, \tau, \beta) = \begin{pmatrix} I(k, k) & I(k, \tau) & I(k, \beta') \\ I(\tau, k) & I(\tau, \tau) & I(\tau, \beta') \\ I(\beta, k) & I(\beta, \tau) & I(\beta, \beta') \end{pmatrix}$$

where

$$(6.2) \quad \begin{aligned} I(k, k) &= n \left[k \psi'(k)^2 - \left(\frac{3}{4k} + 1 \right) \psi'(k) + \frac{1}{4k^2} \right], \\ I(k, \tau) &= \frac{n}{2\tau k} (1 - k\psi'(k)), \\ I(k, \beta') &= \frac{n}{\sqrt{k}\tau} \left(k\psi'(k) - 1 - \frac{1}{2k} \right) \bar{x}. \end{aligned}$$

The other elements of $I(k, \tau, \beta)$ are given in (3.6).

The reference prior depends on the ordering and groupings of the parameters. (For details, see Berger and Bernardo, 1992, Ye, 1990, 1992).

With k as parameter of interest we consider the groupings $\{k, (\tau, \beta)\}$ and $\{k, \tau, \beta\}$.

In the first case, $\pi(k, (\tau, \beta)) \propto \pi(\tau, \beta|k)\pi(k)$ where $\pi(\tau, \beta|k) \propto |h_2|^{-\frac{1}{2}}$, and h_2 is the lower right $(p+1) \times (p+1)$ submatrix of $I(k, \tau, \beta)$, and $\pi(k) \propto [I(k, k)]^{\frac{1}{2}}$.

It follows that

$$(6.3) \quad \pi(k, (\tau, \beta)) \propto \frac{1}{\tau^{p+1}} \left[k\psi'(k)^2 - \left(\frac{3}{4k} + 1 \right) \psi'(k) - \frac{1}{4k^2} \right]^{\frac{1}{2}}.$$

In the second case, $\pi(k, \tau, \beta) \propto \pi(\beta|\tau, k)\pi(\tau|k)\pi(k)$ where $\pi(\beta|\tau, k) \propto |h_1|^{-\frac{1}{2}}$, and h_1 is the lower right $p \times p$ submatrix of $I(k, \tau, \beta)$, $\pi(\tau|k) \propto \frac{\sqrt{n}}{\tau} \left(k\psi'(k) + 1 - \frac{1}{k} \right)^{-\frac{1}{2}}$ is the square root of the $(1, 1)$ -th element of h_2^{-1} , and $\pi(k)$ is the $(1, 1)$ -th element of $[I(k, \tau, \beta)]^{-1}$. Then

$$(6.4) \quad \pi(k, \tau, \beta) \propto \frac{1}{\tau} (k^2\psi'(k)^2 - \psi'(k) - 1)^{\frac{1}{2}} (k^2\psi'(k) + k - 1)^{-\frac{1}{2}}.$$

The two "marginal" priors for k are shown in figure 6.1.

The Jeffreys prior is given by

$$(6.5) \quad \pi^J(k, \tau, \beta) \propto |I(k, \tau, \beta)|^{-\frac{1}{2}} \\ \propto \frac{1}{\tau^{p+1}\sqrt{k}} (k^2\psi'(k)^2 - \psi'(k) - 1)^{\frac{1}{2}}.$$

The marginal posterior distributions of k for the two examples, using prior (6.4), are shown in figure 6.2.

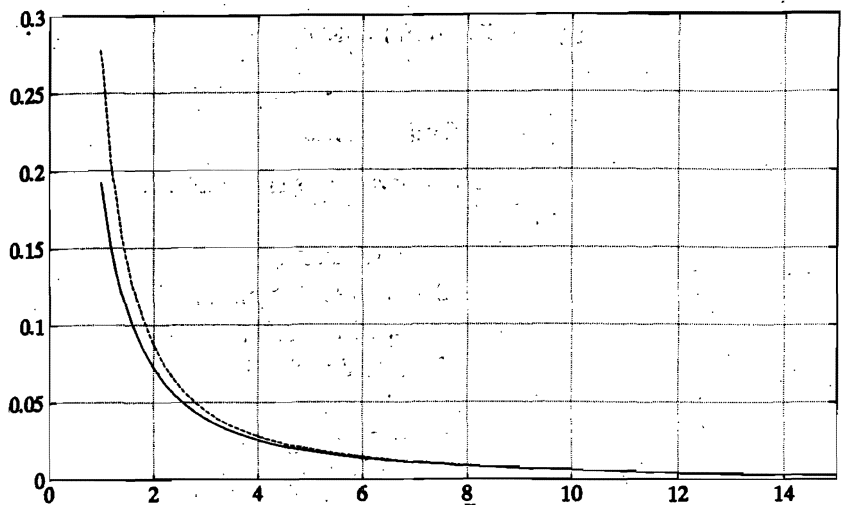


Figure 6.1. Reference priors: $\pi(k, (\tau, \beta))$ (---) and $\pi(k, \tau, \beta)$ (—).

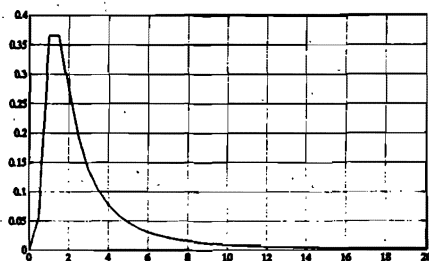
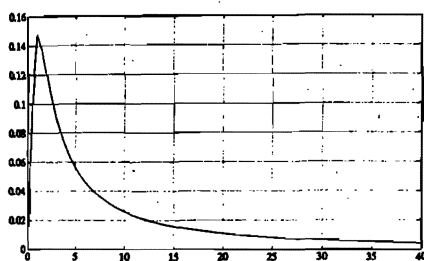


Figure 6.2 Marginal posterior distribution of k for
(a) the Gesell data, and
(b) the Farr data.

The posterior moments of k are

$$E(k) = 10.38, \text{ Var}(k) = 132.09$$

for the Gesell data, and

$$E(k) = 3.68, \text{ Var}(k) = 25.10$$

for the Farr data, suggesting that the deviation from normality is more evident in the second example.

The marginal posterior distributions of the parameters for the Farr data are similar to those in figure 5.5 for $k = 2$ and are not reproduced here. The moments of the marginal posteriors are shown in table 6.1 together with the maximum likelihood estimators under the GEV model.

Table 6.1. Bayes estimates and MLEs of parameters for Farr data

	β_0	β_1	β_2	τ	k
E	-0.9386	0.0439	-0.0491	0.2047	3.68
Var	0.0036	8.5×10^{-6}	1.6×10^{-5}	3.5×10^{-4}	25.10
MLE	-0.9338	0.0440	-0.0496	0.2030	1.97

Comparison with table 5.4 shows the comparatively small difference between the unconditional posterior means and the posterior means given $k = 2$.

For the predictive density of Z_{n+1} given $x_0 = (1 \ 15 \ 8)$ follows

$$E[Z_{n+1}|x_0] = 2.0069, \text{ Var}[Z_{n+1}|x_0] = 0.2106,$$

with 95% prediction interval

$$C = (1.188; 3.031),$$

compared to those given in example 2.

Remarks:

- (i) Of the two examples considered, the deviation from normality appears to be more pronounced for the Farr data. As a subjective rule of thumb, we suggest that a posterior expectation of k larger than 5, as in the case of the Gesell data, would indicate very little loss by using the standard, and that normal assumptions should be adequate. This can be seen from table 5.3.
- (ii) A reparameterization of the perturbation parameter to $K = \frac{1}{k}$ might be a better measure of the perturbation from the standard. However, we have stayed with the present parameterization for convenience of simulation and accurate evaluation of the bigamma and trigamma functions.
- (iii) From table 6.1 it can be seen that there is very little difference between the Bayes estimates and MLEs for the first four parameters. This is because the posterior distributions are almost symmetric. For k , however, the marginal posterior distribution is highly skewed, causing the substantial difference between the MLE and posterior mean. The posterior mode is close to the MLE for k .

The MLEs were evaluated numerically, so it was not possible to calculate the variances.

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APPENDIX

FARR DATA

Z	X_1	X_2	Z	X_1	X_2
2.1156	4	4.6	1.6822	22	9.0
2.1386	4	4.6	1.5933	20	11.0
2.8428	4	4.6	1.6753	20	11.0
1.6239	6	4.6	2.4597	15	9.0
1.6281	6	4.6	2.4975	15	9.0
3.8569	15	20.5	2.8444	10	12.6
3.8572	15	20.5	2.7556	10	12.6
1.4686	17	9.5	3.5647	14	14.2
1.4744	17	9.5	3.5267	14	14.2
2.9839	8	17.0	3.5072	14	14.2
1.1592	22	4.3	2.5906	15	13.0
1.1975	22	4.3	3.4361	18	20.0
3.8444	4	4.5	3.5667	18	20.0
4.1653	4	4.5	2.2189	11	8.7
1.7706	12	8.0	2.2331	11	8.7
1.3772	15	5.8	2.2117	11	8.7
1.9000	8	5.8	2.6575	12	12.5
1.9667	12	5.5	3.1000	12	12.5
1.3814	22	8.5	2.5561	12	12.5
1.4722	22	8.5	5.4014	5	16.0
1.3608	13	8.5	5.4106	5	16.0
2.3417	18	11.0	5.5158	5	16.0
2.3111	18	11.0	2.8622	22	12.5
2.4414	18	11.0	2.4428	22	12.5
2.3822	20	25.0	4.1036	4	12.0
2.4633	20	25.0	3.3833	6	9.1
1.9278	21	8.0	3.3889	6	9.1
1.6100	25	11.0	3.3347	6	9.1
0.8686	15	8.0	3.6319	10	13.0
1.4468	25	9.0	3.7172	10	13.0
1.4675	25	9.0	3.7789	10	13.0
1.4731	22	9.0	5.1489	15	26.0
			4.9736	15	26.0
			4.7236	15	26.0